

Model Analysis Report

Timeseries Statistics Model

MKM-TS-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Flood Alert exceedance sensitivity to threshold level (20-year synthetic record).

Alert Threshold (m)	Total Days	Days/Year
2.50	1120	56.0
2.75	361	18.1
3.00	68	3.40
3.25	10	0.5000
3.50	0	0.0000

Table 2: Summary statistics from 20-year synthetic water level record.

Statistic	Value (m)
Mean level	1.998
Median level	1.998
Std dev	0.466
Min level	0.647
Max level	3.434
P5	1.246
P95	2.747
P99	2.994
Annual maxima mean	3.239
Annual maxima std	0.140

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 3: Stress testing scenarios for Timeseries Statistics Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 4: Performance metrics for Timeseries Statistics Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.